ECON 0100 | Exercise E2 | Competition

Use the animation's competitive firm: $TC = 1000 + q^2$ and $MC = 2q$. Costs include opportunity costs. Answer alone, then compare with a neighbor.

Q1 | Choosing Output

a) The market price is 90. Find total revenue from selling 20 units and from selling 21 units. What is marginal revenue? Explain why it equals the price.

b) At that price, how much should the firm produce? Explain why it should increase output from 40 units but decrease output from 50 units.

c) Calculate the firm's profit at its chosen quantity.

Q2 | Operating or Leaving

At a price of 40, the best positive output is 20 units. Fixed cost is unavoidable this period but avoidable in future periods by leaving the industry.

a) Calculate profit from operating and from shutting down this period. Which should the firm choose?

b) If this price persists, should the firm stay in the industry in the long run? Explain why this answer can differ from (a).

c) Return to the profitable price of 90 in Q1. Would firms enter or exit? What happens to market supply and price?

Q3 | A Higher Fixed Cost

Start again, with the market in long-run equilibrium and every firm earning zero economic profit. Firms have the original costs above. Firms are identical, entry and exit are free, and market demand slopes down.

Every firm must now pay an additional fixed expense of 600 each period, regardless of its output. This period's payment is unavoidable; future payments can be avoided by exiting. Demand and input prices do not change.

a) On the graphs, show the initial equilibrium and the immediate effect of the expense on costs. What happens immediately to price, each firm's output, and profit?

b) Show the adjustment as firms enter or exit. Compared with before the expense, what happens in the long run to price, each remaining firm's output, total market output, and the number of firms? What stops the adjustment?

The Market

One Firm

c) Explain how fewer firms can each produce more while the market produces less in total.